CSP-203: Software Tools & Technologies Laboratory



Lab Session - 6

Introduction to Scripting and Loops

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Objectives:

- ❑ Different special variables
- ❑ Evaluation of expressions
- ❑ Conditional Statement
- ❑ For loop
- ❑ While loop
- ❑ Do while loop
- ❑ Switch case

❑ Special Variables:

```
echo "First argument: $0"
echo "Second argument: $1"
echo "Third argument: $2"
echo "Fourth argument: $3"
echo "Fifth argument: $4"
echo "All arguments: $@"
echo "Number of arguments given to the executable: $#"
echo "Exit status of the last command: $?"
echo "Process number of the current shell: $$"
echo "All the arguments: $*"
~
```

```
arindam@NAMS-LAB-NTN-401-System-1:~/lab-5$ vim script6.sh
arindam@NAMS-LAB-NTN-401-System-1:~/lab-5$ chmod +x script6.sh
arindam@NAMS-LAB-NTN-401-System-1:~/lab-5$ ./script6.sh 1 2 3 4 5
First argument: ./script6.sh
Second argument: 1
Third argument: 2
Fourth argument: 3
Fifth argument: 4
All arguments: 1 2 3 4 5
Number of arguments given to the executable: 5
Exit status of the last command: 0
Process number of the current shell: 3364651
All the arguments: 1 2 3 4 5
arindam@NAMS-LAB-NTN-401-System-1:~/lab-5$
```

❑ Different methods of evaluating and expression:

```
read n
let x=$n+1
echo "$x"
~
```

```
read n
x=$((n+1))
echo "$x"
~
```

```
read n m
x=`expr '(' $n + 1 ')' / $m`
echo "$x"
~
```

```
read n
x=`expr $n + 10`
echo "$x"
~
```

```
read n m
x=`expr \( $n + 1 \) / $m`
echo "$x"
```

- **Conditional Statements:**

```
If [ expression ]
then
  Statement 1
elif [ expression ]
then
  Statement 2
else
  Statement 3
fi
```

- **Test Expression Evaluation:**
  - ✔ n1 -eq n2 : tests if n1 == n2
  - ✔ n1 -ne n2 : tests if n1 != n2
  - ✔ n1 -lt n2 : tests if n1 < n2
  - ✔ n1 -le n2 : tests if n1 <= n2
  - ✔ n1 -gt n2 : test if n1 > n2
  - ✔ n1 -ge n2 : test if n1 >= n2
  - ✔ if either n1 or n2 is not a number then test fails.

arindam@NAMS-LAB-NTN-401-System-1: ~/Desktop/Arindam Thesis/Simulation S

```
echo -n "Enter your age: "
read n
if [ $n -le 10 ]
then
        echo "This child is not allowed to drive."
elif [ $n -le 18 ]
then
        echo "This child is allowd to use bicycle."
else
        echo "He is allowed to drive car."
fi
~
```

```
arindam@NAMS-LAB-NTN-401-System-1:~/lab-5$ vim script5.sh
arindam@NAMS-LAB-NTN-401-System-1:~/lab-5$ chmod +x script5.sh
arindam@NAMS-LAB-NTN-401-System-1:~/lab-5$ ./script5.sh
Enter your age: 17
This child is allowd to use bicycle.
arindam@NAMS-LAB-NTN-401-System-1:~/lab-5$
```

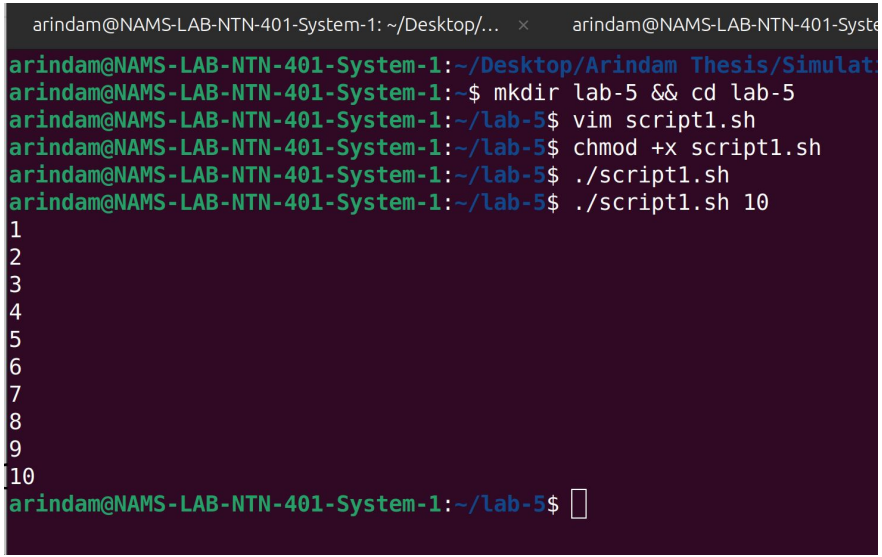
- **For Loop syntax:**

```
for (( initialization; termination condition; update ))
do
  Statement
done
```

Output:
1
2
3
.
.
.
N

- **Example of for loop:**

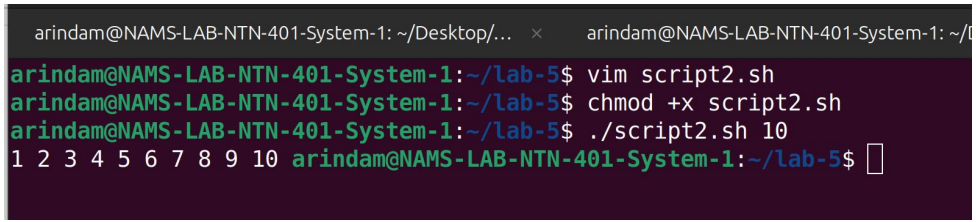
```
N=$1 #read N can also be performed
for (( i = 1; i <= N; i++ ))
do
  echo “$i”
done
```



Output:
1 2 3 … N

- **Example of for loop:**

```
N=$1 # read N can also be performed
for (( i = 1; i <= N; i++ ))
do
  echo –n “$i ”
done
```



- **Example of for loop (Alternative Approach):**

```
N=$1 # read N can also be performed
for i in $(seq 1 $N)
do
  echo –n “$i ”
done
```

Output:
1 2 3 … N

```
arindam@NAMS-LAB-NTN-401-System-1:~/lab-5$ vim script3.sh
arindam@NAMS-LAB-NTN-401-System-1:~/lab-5$ chmod +x script3.sh
arindam@NAMS-LAB-NTN-401-System-1:~/lab-5$ ./script3.sh 10
12345678910arindam@NAMS-LAB-NTN-401-System-1:~/lab-5$
```

- **Printing list of given arguments:**

```
for i in $1 $2 $3 $4 $5
do
  echo –n “$i ”
done
```

Output:
Argument1 argument2 … arguement3

```
arindam@NAMS-LAB-NTN-401-System-1:~/lab-5$ vim script4.sh
arindam@NAMS-LAB-NTN-401-System-1:~/lab-5$ chmod +x script4.sh
arindam@NAMS-LAB-NTN-401-System-1:~/lab-5$ ./script4.sh 5 4 10 20
541020arindam@NAMS-LAB-NTN-401-System-1:~/lab-5$
```

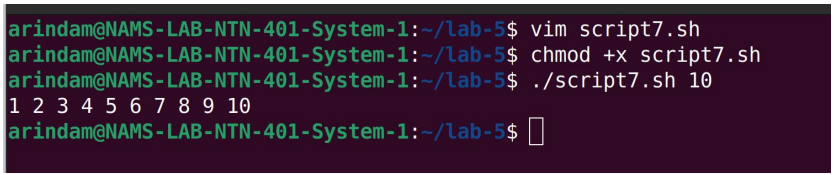
- **While loop syntax:**

```
while [ condition ]
do
  Statement
done
```

- **Example of while loop:**

```
N=$1 # read N can also be performed
i=1
while [ $i -le $N ]
do
  echo -n “$i ”
  ((i++))
done
echo
```

Output:
1 2 3 … N



- **Example of while loop:**

```
N=$1 #read N can also be performed
i=1
while true
do
  echo -n "$i ”
  ((i++))
  if [ "$i" -gt "$N" ]
  then
        break
  fi
done
echo
```

Output:
1 2 3 … N

```
arindam@NAMS-LAB-NTN-401-System-1:~/lab-5$ chmod +x script8.sh
arindam@NAMS-LAB-NTN-401-System-1:~/lab-5$ ./script8.sh 8
12345678
arindam@NAMS-LAB-NTN-401-System-1:~/lab-5$
```

- **Switch case:**

```
echo -n "Enter the choice(1-4): “
read n
case $n in
  1)
        echo "Statement 1“
        ;;
  2)
        echo "Statement 2“
        ;;
  3)
        echo "Statement 3“
        ;;
  4)
        echo "Statement 4“
        ;;
  *)
        echo "Default Statement“
        ;;
esac
```

Input: 2
Output: Statement2

Input: 8
Output: Default Statement

- **Switch case:**

```
echo -n "Enter the choice(1-4): “
read n
case $n in
  1)
        echo "Statement 1“
        ;;
  2)
        echo "Statement 2“
        ;&
  3)
        echo "Statement 3“
        ;;
  4)
        echo "Statement 4“
        ;;
  *)
        echo "Default Statement“
        ;;
esac
```

```
arindam@NAMS-LAB-NTN-401-System-1:~/lab-5$ chmod +x script9.sh
arindam@NAMS-LAB-NTN-401-System-1:~/lab-5$ ./script9.sh
Enter the number: 2
Statement 2
Statement 3
arindam@NAMS-LAB-NTN-401-System-1:~/lab-5$ ./script9.sh
Enter the number: 1
Statement 1
arindam@NAMS-LAB-NTN-401-System-1:~/lab-5$ ./script9.sh
Enter the number: 10
arindam@NAMS-LAB-NTN-401-System-1:~/lab-5$
```

Input: 2
Output: Statement2
Statement3

- **Switch case:**

```
echo -n "Enter the choice(1-4): “
read n
case $n in
  1)
        echo "Statement 1“
        ;;
  2)
        echo "Statement 2“
        ;&
  3)
        echo "Statement 3“
        ;;
  “NAMS”)
        echo "Statement 4“
        ;;
  *)
        echo "Default Statement“
        ;;
esac
```

```
arindam@NAMS-LAB-NTN-401-System-1:~/lab-5$ ./script9.sh
Enter the number: NAMS
Statement 4
arindam@NAMS-LAB-NTN-401-System-1:~/lab-5$ ./script9.sh
Enter the number: nams
Default Statement
```

Input: 2
Output: Statement2
Statement3

❑ String comparison:

```
echo "Enter the string 1: "
read string1
echo "Enter the string 2: "
read string2
if [ "$string1" == "$string2" ]
then
        echo "Same"
else
        echo "Different"
fi
```

```
arindam@NAMS-LAB-NTN-401-System-1:~/lab-5$ ./script1.sh
Enter the string 1:
IITBhilai
Enter the string 2:
IITBhilai
Same
arindam@NAMS-LAB-NTN-401-System-1:~/lab-5$ ./script1.sh
Enter the string 1:
iitBhilai
Enter the string 2:
IITBhilai
Different
arindam@NAMS-LAB-NTN-401-System-1:~/lab-5$
```

❑ String comparison:

```
read -p "Enter string1: " string1
read -p "Enter string2: " string2
if [ "$string1" == "$string2" ]
then
        echo "Same"
else
        echo "Different"
fi
~
```

```
arindam@NAMS-LAB-NTN-401-System-1:~/lab-5$ vim script1.sh
arindam@NAMS-LAB-NTN-401-System-1:~/lab-5$ ./script1.sh
Enter string1: IIT Bhilai Campus
Enter string2: IIT Bhilai Campus
Same
arindam@NAMS-LAB-NTN-401-System-1:~/lab-5$ ./script1.sh
Enter string1: IIT Bhilai campus
Enter string2: IIT Bhilai Campus
Different
arindam@NAMS-LAB-NTN-401-System-1:~/lab-5$
```

❑ String comparison (Case in-sensitive):

```
read -p "Enter the string 1: " string1
read -p "Enter the string 2: " string2
if [ "${string1,,}" == "${string2,,}" ]
then
        echo "Same"
else
        echo "Different"
fi
~
~
~
~
```

```
arindam@NAMS-LAB-NTN-401-System-1:~/lab-5$ vim script1.sh
arindam@NAMS-LAB-NTN-401-System-1:~/lab-5$ chmod +x script1.sh
arindam@NAMS-LAB-NTN-401-System-1:~/lab-5$ ./script1.sh
Enter the string 1: IIT Bhilai Campus
Enter the string 2: iit Bhilai campus
Same
arindam@NAMS-LAB-NTN-401-System-1:~/lab-5$ ./script1.sh
Enter the string 1: IIT BHilai
Enter the string 2: IIT bhilai campus
Different
arindam@NAMS-LAB-NTN-401-System-1:~/lab-5$
```


THANK
YOU